MIL-C-81064(OS)
AMENDMENT 7
30 March 1971
Superseding
AMENDMENT 6
29 January 1971

MILITARY SPECIFICATION

COURSE GYROSCOPE

(For Torpedo Mk 46 Mods 0 and 1)

This amendment forms a part of Military Specification MIL-C-81064(OS) of 21 May 1964 and has been approved by the Naval Ordnance Systems Command, Department of the Navy.

By this change, basic specification MIL-C-81064(Wep) is changed to MIL-C-81064(OS).

Wherever it appears in specification delete "Bureau of Naval Weapons" and substitute "Naval Ordnance Systems Command"; delete "BUWEPS" and substitute "NAVORD"; delete the term "Acceptance test" and substitute the term "Quality conformance test."

Page 1: Delete from title "(For Torpedo Mk 46 Mod 0)" and substitute "(For Torpedo Mk 46 Mods 0 and 1)."

Page 5, paragraph 3.3.1.4. Delete and substitute:
"3.3.1.4 Firing modes

3.3.1.4.1 Squib fired units. Uncaging shall be accomplished by application of 1 ampere dc minimum for 15 milliseconds minimum. Uncaging circuitry before firing shall present a 2 ±0.7 ohm load.

3.3.1.4.2 Solenoid fired units. The solenoid having a resistance between 1.3 to 50 ohms shall be capable of spring release and gimbal uncage in any of the following modes of operation:

(a) By applying 19 to 30 vdc or 31 to 35 vrms, 60 Hz, for 50 to 90 ms to an external circuit consisting of an RC32GF8R2J resistor and a JAN1N645 diode connected in series with gyro terminal P. The return line for this voltage shall be connected to gyro terminal L.

(b) By applying 33 to 35 vdc for 20 to 30 ms to an external circuit consisting of an M22684-02-0015 (20Ω) resistor and a JAN1N645 diode connected in series with gyro terminal P. The return line for this voltage shall be connected to gyro terminal L."

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Page 5, paragraph 3.3.1.5. Delete and substitute:
"3.3.1.5 Insulation resistance. The gyro insulation resistance shall be 50 megohms minimum at 200 vdc for 15 seconds between isolated gyro circuits and between these circuits and the case."

Page 5, paragraph 3.3.1.6. Delete and substitute:
"3.3.1.6 Wound and caged indications. The gyro shall have a positive means of indicating wound and caged condition. The indicating circuit shall have a resistance of 5.0 ohms maximum when measured with current of 7.5 ma maximum."

Page 5, paragraph 3.3.2.1: Delete "a maximum defined by one or the other of."

Page 5, paragraph 3.3.2.1(b): Delete "A 7.5 degree Scorsby drift rate" and substitute "The gyro drift rate with 7.5 degree Scorsby action."

Page 5, paragraph 3.3.2.2. Delete and substitute:
"3.3.2.2 Scorsby-action drift rate. The gyro drift with 7.5 degree Scorsby action, with only the spring motor operating, and with the gyro mounted in any direction on the Scorsby table shall not exceed 1.5 degrees at any time during the first 30 seconds after uncaging."

Page 7, Table 1: Delete "minimum" from "Double Amplitude" and "Acceleration" column; add "Minimum" after "0.75", "0.1", "0.036", "1.3", "3.0", and "10.0."

Page 9, Table IV: Change entries in right column to read:
"-45 to +140 degrees F
0 to +140 degrees F
0 to +160 degrees F"

Page 9, paragraph 4.1. Delete and substitute:
"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

Pages 10 and 11. Delete paragraphs 4.3.2 and 4.3.2.1 and substitute:
"4.3.2 Periodic production sampling. Periodic production samples

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shall be selected from units that have passed the quality conformance inspections of 4.7 satisfactorily. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production, select at random 1 unit from each group of 30 units. Upon satisfactory compliance of samples selected from 3 consecutive groups shift to Step 2. Each sample so selected shall be numbered consecutively, starting with sample Number 1.

Step 2. Select at random 1 unit from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is found. Each sample so selected shall be numbered consecutively, starting with a sample number one greater than the highest sample number selected in Step 1.

4.3.2.1 Defective units. When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

(a) Determine the cause of the defect.

(b) Take the necessary action to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, eliminate defects from those previously shipped.

(c) Submit to the procuring activity for approval the proposed corrective action to prevent recurrence of the defect.

(d) Submit to the procuring activity for approval the proposed retest program to verify that the corrective action eliminated the defect.

(e) Return to Step 1 and proceed as indicated when inspection is resumed."

Page 12, Table V: Delete heading "Computer Number" and substitute "Gyro Number."

Page 12, paragraph 4.5.1.4: Delete "over a range of -45 to +140 degrees F" and substitute "over a range of -50 to +145 degrees F."

Page 12, paragraph 4.5.1.5: Delete "at 100 degrees F" and substitute "at 110 degrees F."

Page 13, paragraph 4.5.2.1.2: In lines 5 and 6 delete repeated sentence that reads "The log frequency sweep between 500 and 5 cycles shall be conducted in 7.5 minutes."

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Page 15, paragraph 4.5.2.4.2. Delete and substitute:
"4.5.2.4.2 Temperature, operating. The gyro shall be stabilized at zero degrees F and tested to conform to the quality conformance test requirements of 4.7. The gyro shall then be stabilized at +160 degrees F and tested to conform to the quality conformance test requirements of 4.7."

Page 15, paragraph 4.5.2.5. Delete and substitute:
"4.5.2.5 Pressure test. The gyro shall be subjected to a pressure of 2.7 psia. The gyro shall meet the quality conformance test requirements specified in 4.7 after the pressure test."

Page 16, paragraph 4.5.2.7. Delete and substitute:
"4.5.2.7 Life test. The gyro shall be operated through 25 operating cycles (see 6.3.7). The gyro shall be capable of meeting the quality conformance tests requirements specified in 4.7 during and at the conclusion of the life test. The operating cycles accumulated during environmental tests shall be included in the required 25 cycles."

Page 16, paragraphs 4.6, 4.6.1, 4.6.1(a), and 4.6.1(b). Delete and substitute:
"4.6 Periodic production tests. The periodic production sample, selected in accordance with 4.3.2, shall be subjected to the tests of Table X in the order specified.

Table X

Periodic Production Test Schedule

Test	Sample and Test Sequence		
	1, 4, etc.	2, 5, etc.	3, 6, etc.
Vibration (4.5.2.1)	1*		1*
Acceleration (4.5.2.3)	2*	1*	
Shock (4.5.2.2)	3*	2*	
Temperature, Nonoperating (4.5.2.4.1)	4		3
Temperature, Operating (4.5.2.4.2)	5	3	
Pressure (4.5.2.5)	6		2
Humidity (4.5.2.6)	7	4	
Life (4.5.2.7)	8		
Quality conformance (4.7)	9	5	4

*Visual Check after Testing

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4.6.1 Periodic production test equipment. The items of test equipment shall be those specified for conducting the tests of Sections 4.5 and 4.7."

Page 16, paragraph 4.6.2: Delete in its entirety.

Preparing Activity
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